

Department of Obstetrics and Gynecology
National Clinical Research Center for Obstetrics and Gynecology
Tongji Hospital, Tongji Medical College
Huazhong University of Science and Technology
Wuhan 430030, China
E-mail: zhutao@tjh.tjmu.edu.cn

August 1, 2026

The Editor-in-Chief
Briefings in Bioinformatics
Oxford University Press

Dear Editor,

We are pleased to submit our manuscript entitled “**Complementary Information Channels for Synthetic Lethality Prediction: Augmenting Graph Topology with Protein Language Models and LLM Weak Supervision**” for consideration for publication in *Briefings in Bioinformatics*.

Summary of the Work

Synthetic lethality (SL) prediction is fundamental to anticancer drug target discovery, yet cold-start generalization—predicting SL for gene pairs unseen during training—remains challenging for all existing methods. We present a three-layer collaborative framework that integrates large language model (LLM) literature mining, GATv2 graph neural networks, and ESM-2 protein language model embeddings. Through rigorous head-to-head comparison, we demonstrate that while PPI topology is the dominant information source for SL prediction, our orthogonal information channels—ESM-2 embeddings (+12.2% AUC) and LLM weak supervision (+1.6% AUC)—provide complementary predictive signals.

Key Contributions and Novelty

Our work makes three distinct contributions to the SL prediction methodology literature:

1. **Identification and quantification of SL adjacency information leakage** in standard multi-view GAE cold-start evaluations: we demonstrate that including test SL pairs in the adjacency matrix artificially inflates CV3 AUC by 13.7%. We propose fair masking as a required protocol for future benchmark studies, extending the Feng *et al.* (2024, *Nat. Commun.*) benchmark framework.
2. **Discovery and quantification of zero-vector ablation bias** in batch-normalized GNNs: we show that the conventional practice of using zero vectors for feature ablation overestimates feature importance by $14\times$ (apparent +22.9% vs. true +1.6%). We recommend shuffled-feature ablation as a new methodological standard.
3. **The first demonstration of LLM weak supervision as a literature mining paradigm for SL prediction**, achieving 100% NER precision at <\$5 total API cost, and validated through independent DepMap CRISPR screening across 1,208 cell lines ($1.73\times$ enrich-

ment, Fisher's exact $p=0.041$).

Relevance to *Briefings in Bioinformatics*

Our manuscript aligns with the journal's scope across multiple dimensions: (i) it presents a novel computational methodology integrating three rapidly converging trends (GNNs, protein language models, and LLM-based literature mining); (ii) it provides two broadly applicable methodological insights for evaluation protocol design in biomedical machine learning; and (iii) it includes comprehensive independent validation through DepMap CRISPR data. The honest, self-critical evaluation style—acknowledging the dominance of graph topology over our complementary signals—reflects the methodological rigor expected by *Briefings in Bioinformatics*.

Additional Information

This manuscript is a single-author work. The author declares no competing interests. All data sources are publicly available; code will be deposited in a public GitHub repository upon acceptance. This manuscript has not been published or submitted elsewhere.

We believe our work will be of significant interest to the *Briefings in Bioinformatics* readership, particularly computational biologists working on network-based prediction, protein language model applications, and biomedical text mining. We look forward to your consideration.

Sincerely,

Tao Zhu, M.D.